# Parallel Programming: Now What?


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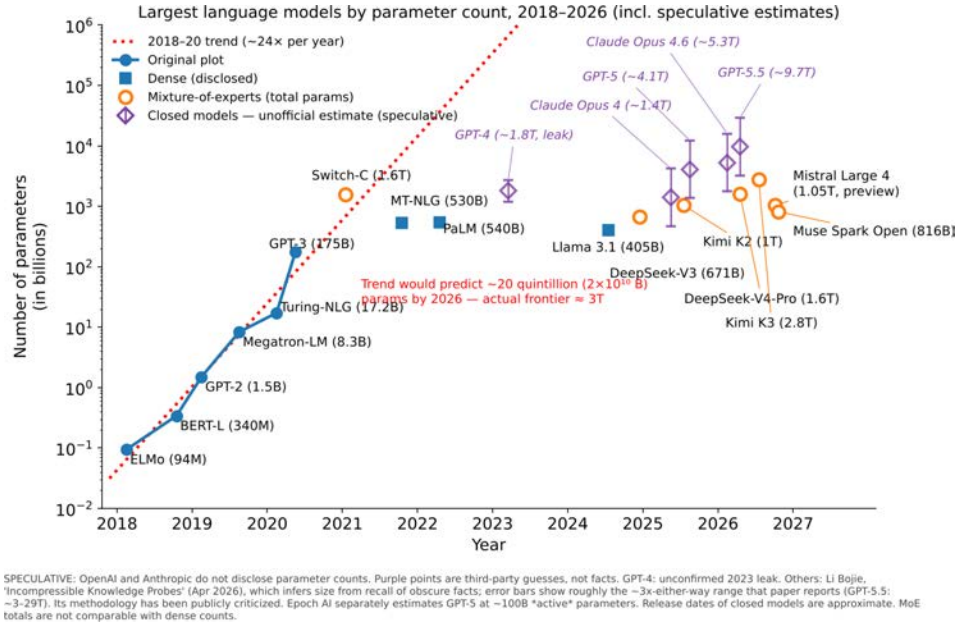
**So what?**
**Learned the current (MPI+OpenMP+CUDA) & emerging (MPI+OpenMP target/SYCL) parallel programming languages**

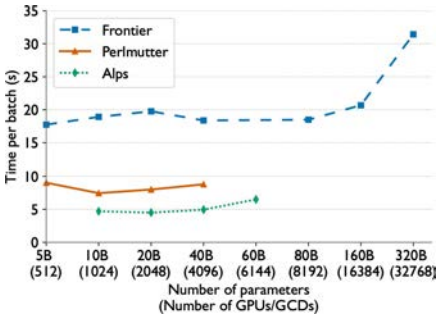
**Now what?**



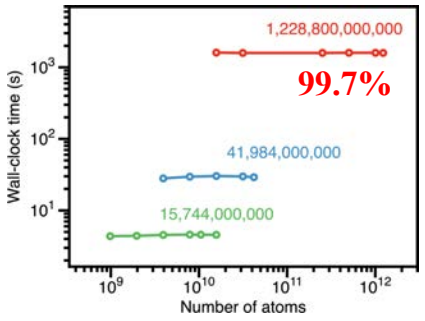
# Extreme-Scale AI

- **Trillion-parameter large language models (LLMs) have been trained on 10,000+ GPUs**




[Democratizing AI: open-source scalable LLM training on GPU-based supercomputers](), Singh *et al.*, *SC24*






**Test-drive and profile open-weight/source LLMs for final project?**

***cf.* Nearly perfect weak scalability of foundation material AI model on 60,000 GPUs**

[Multiscale light-matter dynamics in quantum materials](), T. M. Razakh *et al.*, *2025 Gordon Bell prize finalist*

# Google Tensor Processing Unit

- **Google's tensor processing unit (TPU) accelerators are available on cloud**
- **XLA (accelerated linear algebra) is a compiler for TensorFlow applications on TPU**

https://cloud.google.com/tpu

- **For physics-informed AI, use JAX software built on Autograd (automatic differentiation)—both on GPU & TPU**

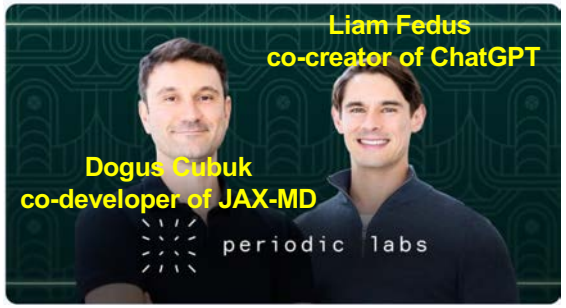
w.r.t. model parameters

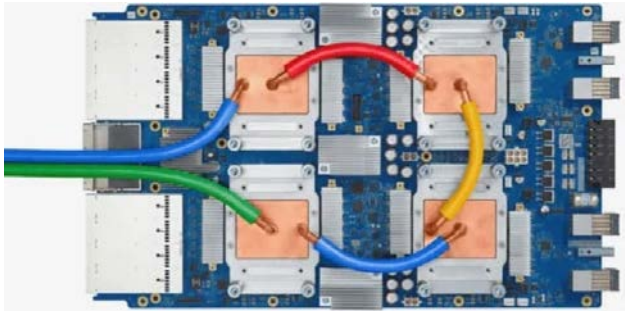
https://github.com/google/jax

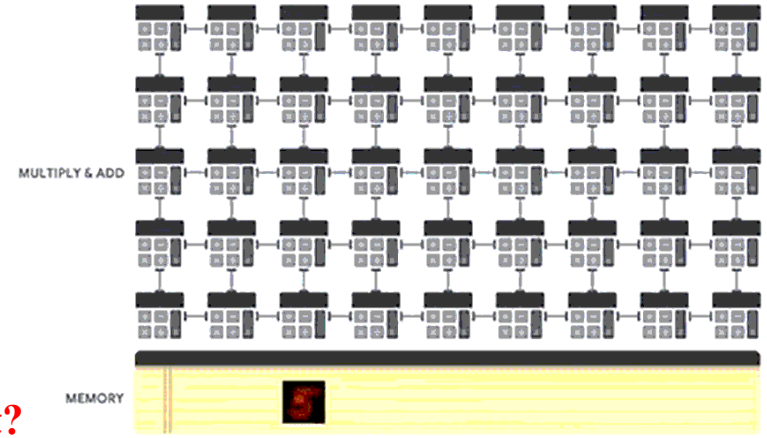
- **JAX-MD is an accelerated, differentiable molecular dynamics engine**

https://github.com/google/jax-md

**Periodic Labs eyes $7B valuation, fivefold jump in six months**



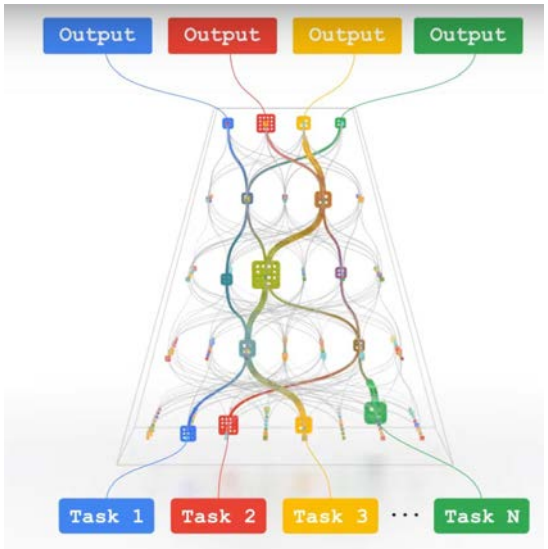






**Test-run & profile for final project?**

# Google's Pathways to AI Future

- **Pathways—a new AI architecture—will handle many tasks at once, learn new tasks quickly and reflect a better understanding of the world for human-like general AI**



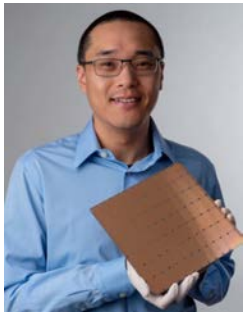

AI isn't as smart as you think — but it could be, Jeff Dean, *TED Talk*
Introducing Pathways: a next-generation AI architecture

*cf.* Sparse mixture-of-experts transformers for efficient scaling of large language models, T. M. Ghazal, *Proc. CS* **275**, 923 ('26)

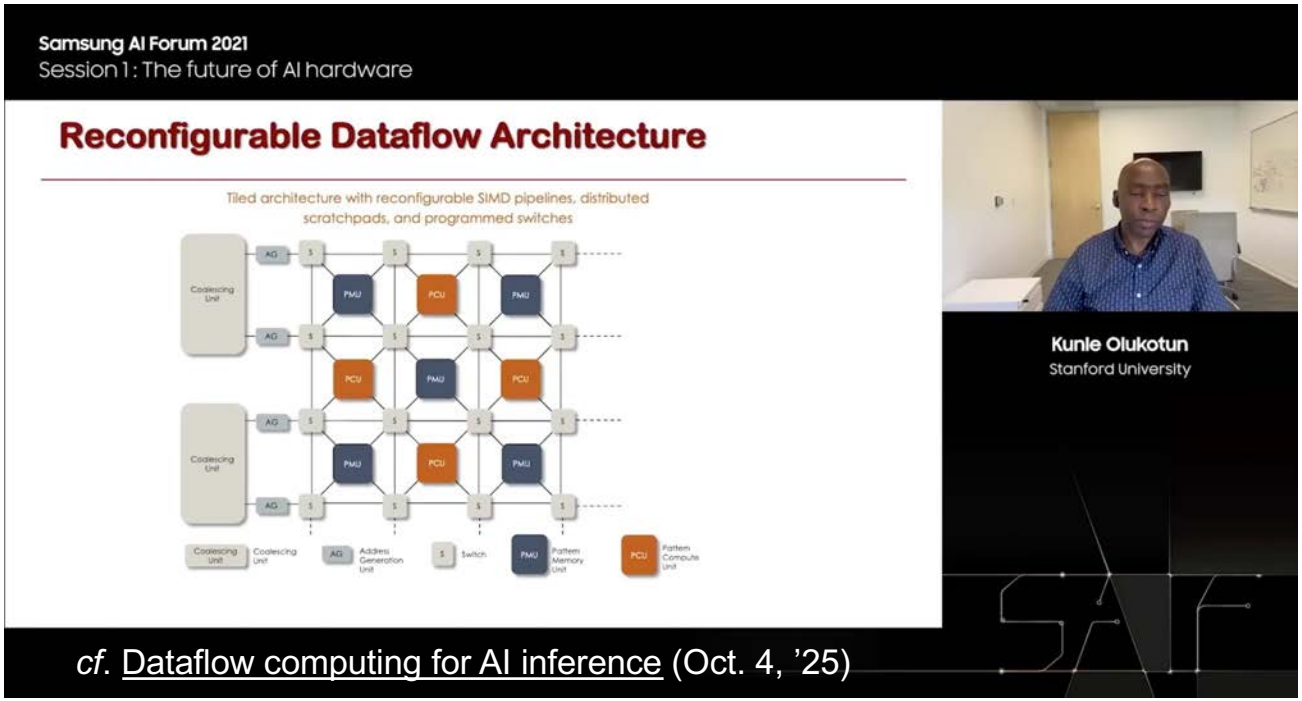
# GPU & TPU Are No Good: 0×0 = 0 Problem

- **It's sparsity: A lot of "multiply by zero" operations degrade speed & power efficiency**

*cf.* "Selectable sparsity" on Cerebras AI chip
https://cerebras.net/



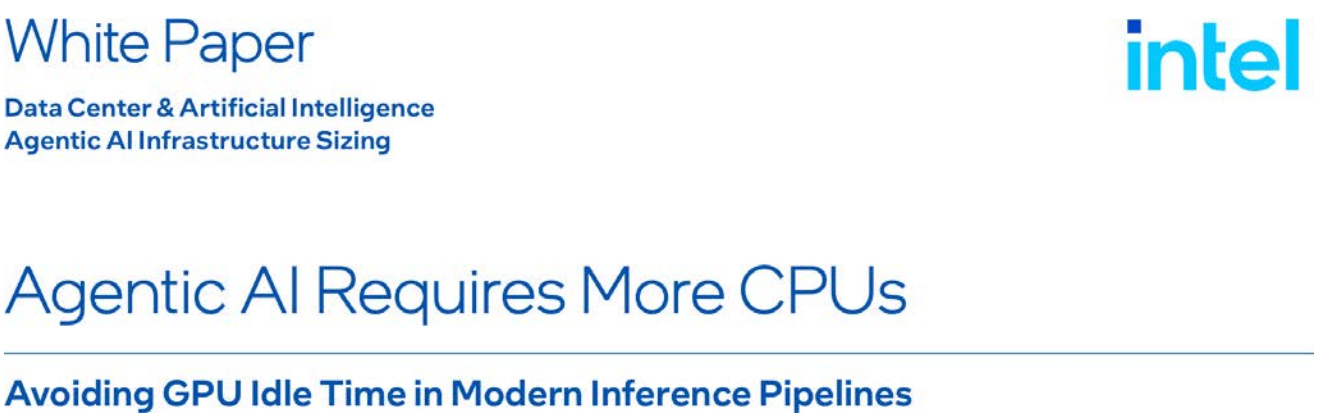
- **Sparse dataflow: Need new architectures & programming models**




*cf.* Slide 13 of OpenMP lecture

*cf.* Dataflow computing for AI inference (Oct. 4, '25)

# Also, Back to CPU: Idle GPU Problem




- ***Agentic AI workflows*** **function like** ***distributed systems*****, introducing** ***CPU-intensive stages*** **like orchestration, reflection & planning, which can slow down the whole process & leave** ***GPUs sitting idle***

  Computation-communication overlap, asynchronicity & latency hiding (*cf*. MPI Irecv function & OpenMP nowait clause)

https://www.intel.com/content/www/us/en/content-details/916705/agentic-ai-requires-more-cpus.html

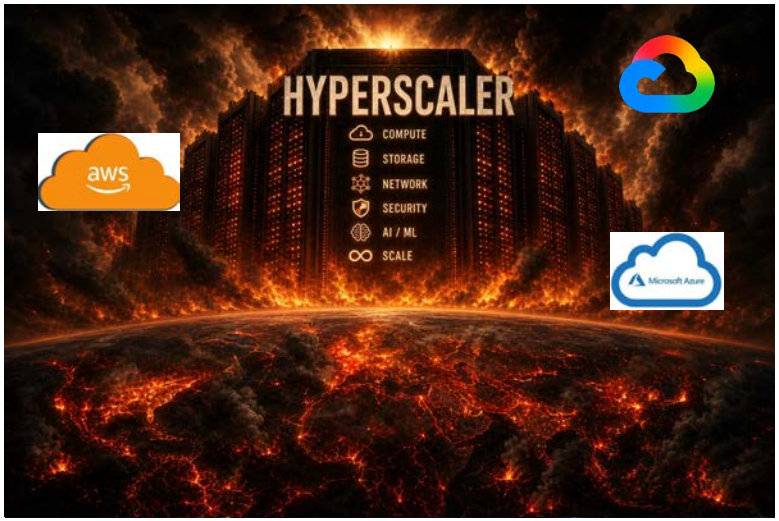
# Will AI Destroy the Earth?

WIRED

REECE ROGERS GEAR JUL 11, 2024 6:30 AM

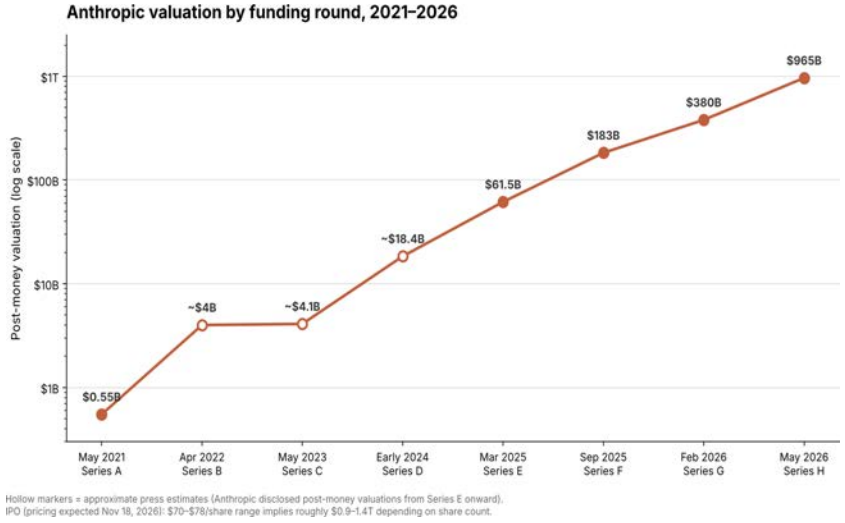
## AI's Energy Demands Are Out of Control. Welcome to the Internet's Hyper-Consumption Era

**Generative artificial intelligence tools, now part of the everyday user experience online, are causing stress on local power grids and mass water evaporation.**







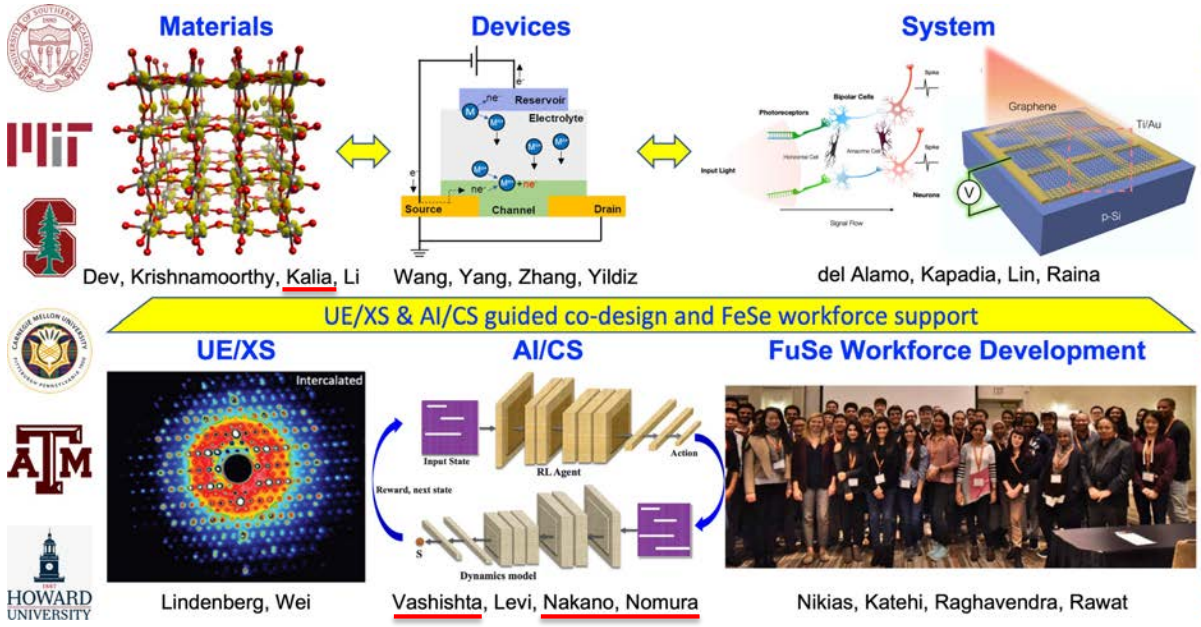



*cf.* Exponential AI-chip circular economy

https://www.wired.com/story/ai-energy-demands-water-impact-internet-hyper-consumption-era/

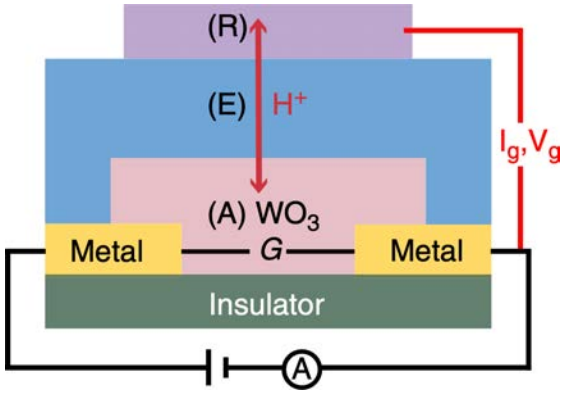
# Solution? Future Semiconductors

- **USC-MIT-Stanford-CMU-TAMU-Howard team received a U.S. National Science Foundation (NSF) Future Semiconductors (FuSe) teaming grant (TG) for atoJoule (aJ) in-sensor computing for sustainable AI future**



# Examples: AttoJoule Neurons

- **Identify atomistic & electronic mechanisms of emerging aJ in-sensor neuromorphic computing without external power:**
  1. **Protonic synapse that is deterministic & high speed with aJ energy consumption**
  2. **Retinal neurons for in-sensor image computing without external power**




**Conductivity switch in H-doped $WO_3$**

Protonic synapse (MIT)

Onen *et al.*, *Science* **377**, 539 ('22)

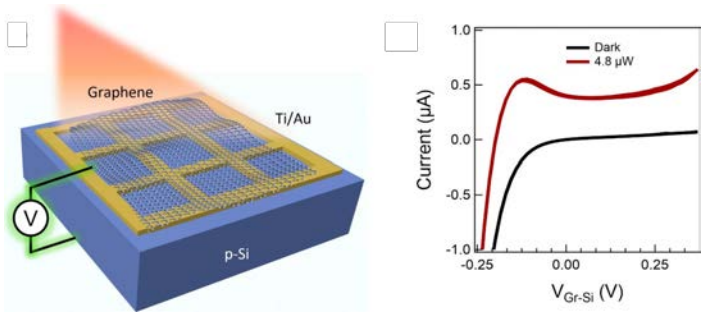



**Photo-induced negative differential resistance (NDR) in graphene/Si**

Oscillatory retinal neural network (USC)

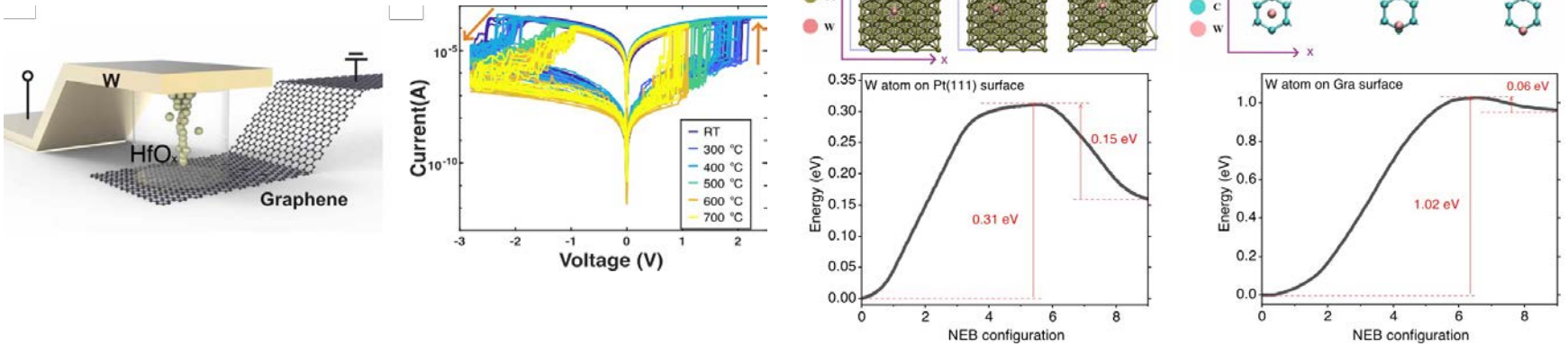
Ahsan *et al.*, *ACS Nano* **18**, 23785 ('24)

Hokyo *et al.*, *J. Phys. Chem. Lett.* **15**, 9226 ('24)

- **Using our breakthrough AI-driven simulation technologies:**
  1. **Nonadiabatic quantum molecular dynamics (NAQMD) to study photoexcitation dynamics involving electrons & nuclei** T. M. Razakh *et al.*, *2025 Gordon Bell Prize finalist*, doi: 10.1145/3712285.3771785
  2. **AI-accelerated neural-network quantum molecular dynamics (NNQMD) with state-of-the-art accuracy, speed & robustness:** Hattori *et al.*, *Nature Commun.* **17**, 8225 ('26) doi: 10.1038/s41467-026-74919-8

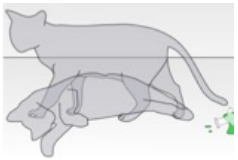
# Go Neural!

- **Prof. Joshua Yang's group at USC developed high-temperature memristors operating above record 700°C based on graphene/$HfO_x$/tungsten**
- **Our first-principles calculations attributed the enhanced thermal stability to weaker tungsten adsorption & higher surface diffusion barriers on graphene**

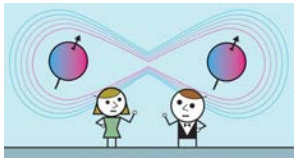



High-temperature memristors enabled by interfacial engineering,
J. Jian *et al.*, *Science* **392**, 771 ('26)

# Or Go Quantum?



**Quantum computing utilizes quantum properties such as superposition & entanglement for computation**



- **Quantum advantage (*i.e.*, quantum computer is faster than the fastest supercomputer) was demonstrated by Google**
  F. Arute, J. M. Martinis *et al.*, *Nature* **574**, 505 ('19)
- **Google's Sycamore quantum computer consumed 26 kilowatts of power to outperform the 13 megawatts Summit supercomputer at Oak Ridge National Lab, *i.e.*, 500 times more energy efficient**



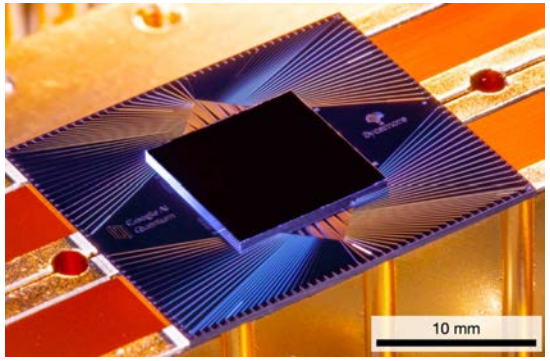

9,216-CPU & 27,648-GPU Summit

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***Fast & frugal***








54-qubit Google Sycamore

# Quantum-Centric Supercomputing



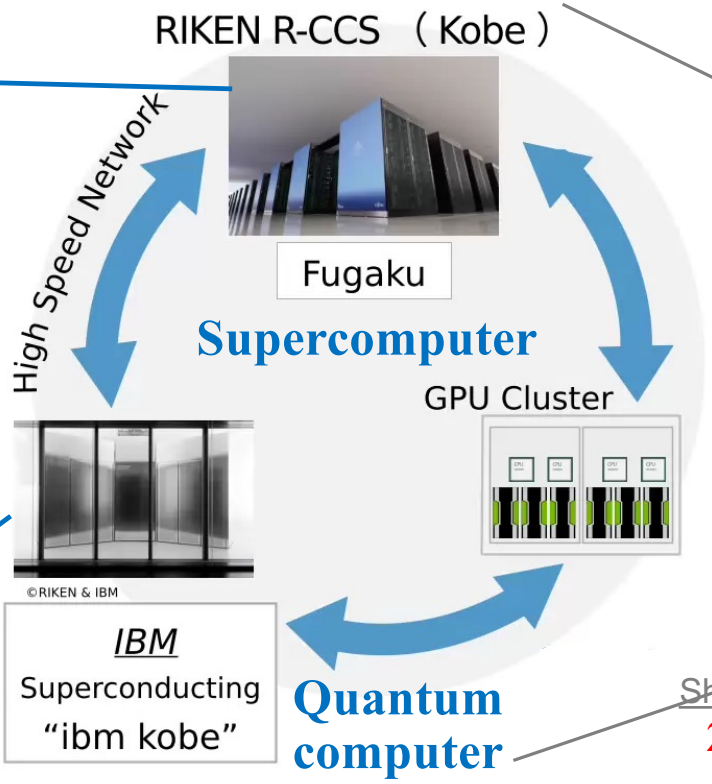
RunClassical()
C++/MPI

System call

RunQuantum()
Python/Qiskit



*Hybrid quantum+supercompuring*










**State-of-the-art QCSC**

Shirakawa *et al.*, *arXiv:* 2511.00224 (’25)

**2026 Gordon Bell Prize finalist**

Merz *et al.*, *arXiv*: 2605.01138v2

- **Many next-generation supercomputers will be tightly integrated with quantum computers as quantum processing unit (QPU), *e.g.*, FugakuNEXT at RIKEN, Japan & NERSC at Berkeley, U.S. in ~2030**

# Computer Science Perspective

## "Intelligent Heuristics Are the Future of Computing"

SHANG-HUA TENG, University of Southern California (USC), USA

Back in 1988, the partial game trees explored by computer chess programs were among the largest search structures in real-world computing. Because the game tree is too large to be fully evaluated, chess programs must make heuristic strategic decisions based on partial information, making it an illustrative subject for teaching AI search. In one of his lectures that year on AI search for games and puzzles, Professor Hans Berliner — a pioneer of computer chess programs[1] — stated:

"*Intelligent heuristics are the future of computing.*"

As a student in the field of the theory of computation, I was naturally perplexed but fascinated by this perspective. I had been trained to believe that "Algorithms and computational complexity theory are the foundation of computer science." However, as it happens, my attempts to understand heuristics in computing have subsequently played a significant role in my career as a theoretical computer scientist. I have come to realize that Berliner's postulation is a far-reaching worldview, particularly in the age of big, rich, complex, and multifaceted data and models, when computing has ubiquitous interactions with science, engineering, humanity, and society. In this article,[2] I will share some of my experiences on the subject of heuristics in computing, presenting examples of theoretical attempts to understand the behavior of heuristics on real data, as well as efforts to design practical heuristics with desirable theoretical characterizations. My hope is that these theoretical insights from past heuristics — such as spectral partitioning, multilevel methods, evolutionary algorithms, and simplex methods — can shed light on and further inspire a deeper understanding of the current and future techniques in AI and data mining.

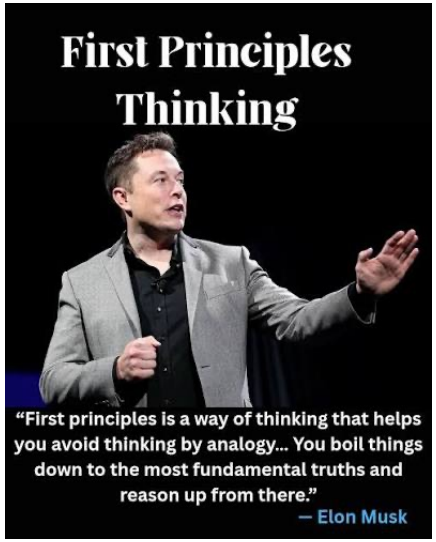


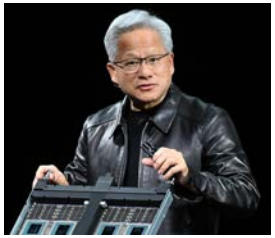

*ACM Trans. Intell. Syst. Technol.* ('23)
https://doi.org/10.1145/3627708

# Age of Physical AI

- **Elon Musk applies *first-principles physics thinking*—breaking complex systems down to fundamental, indisputable truths rather than reasoning by analogy—to guide xAI's Grok AI models**
- **Grok architecture**
  - **> *JAX*: High-performance & auto-differentiable Python library for numerical computing & AI**
  - **> *Sparse mixture-of-experts*: Scalable large language models with data, pipeline & expert parallelism**
  - **> *Physics-informed reinforcement learning* for algorithm updates, *cf*. self-improving AI by Darwin-Gödel machine**

    Zhang *et al*., *ICLR 2026*: doi: 10.48550/arXiv.2505.22954



*"Physics is the law, everything else is a recommendation"*
*"Physical AI could expand the global economy million-fold"*
*"Energy is the new currency"*
Elon Musk

https://finance.biggo.com/news/17504f63-c911-4dcc-a04e-9eb32ed5bc55

- **Elon Musk & Jensen Huang: *Understanding of physics and mathematics will be more decisive* in a world where AI automates software routines**

https://singularity-experts.com/en/blog/leaders-recommend-studying-physics-and-mathematics-instead-of-programming

# Where to Go from Here

- **Expect computing paradigm changes**
- **Stay curious to "understand"**




1976-2026
THOMAS LORD DEPARTMENT OF COMPUTER SCIENCE
50 YEARS. LIMITLESS FUTURES.

USC